DS Foundations Lesson Plan (V2)

📘 Lesson Plan: Data Science Foundations for AI-Driven Systems

How AI Thinks About Data, Models, and Decisions

Position in Curriculum

CS Foundations → Data Science Foundations (this) → Data Contracts & Upstream Risk → Production Foundations

This lesson plan defines *what data science means* when working alongside AI systems — before production constraints fully apply.

Audience

Data scientists, data engineers, and applied researchers collaborating with AI systems to design, evaluate, and reason about analytical pipelines — especially in scientific, policy, or decision-support contexts.

Prerequisite

Completion of **CS Foundations for Agentic & Multimodal AI (V3)**

Why This Lesson Plan Exists

Data science is often described by tools and outputs.
AI does not reason this way.
To AI systems, **data science is a decision discipline**:
transforming uncertainty into constrained, defensible decisions using data.
This lesson plan:

- formalizes how AI reasons about the data science lifecycle,
- reframes pipelines as **decision graphs**, not scripts,
- and prepares practitioners to enter production conversations already aligned with Genesis-style system thinking.

MODULE 0 — What "Data Science" Means to AI

Reframing the Discipline

Why This Matters

Humans often define data science by tooling (pandas, models, dashboards).AI defines data science by **decision structure**.

Without this reframing, downstream modeling and production decisions are misaligned.

Core Mental Model Introduced

To AI, data science is:

Inference under constraints in service of a decision

Pipelines exist to **shape uncertainty**, not just process data.

Key Concepts

- Data science as decision-making under uncertainty
- Pipelines as decision graphs, not linear workflows
- Explicit vs implicit assumptions
- Data ≠ truth; data = evidence with bias

AI Decision Questions

At every stage, AI implicitly asks:

1. What decision is downstream of this?
2. What uncertainty am I reducing?
3. What constraints matter most here?

🧪 Exercise

Take a past data science project and rewrite its goal as a decision statement:

“We are deciding X under Y constraints with Z uncertainty.”

MODULE 1 — Data Ingestion: How AI Thinks About Inputs

Trust Calibration, Not File Loading

Why This Matters

AI treats ingestion as **trust calibration**, not retrieval.

Ingested data sets the ceiling for every downstream decision.

AI Decision Questions

- Is this source authoritative, proxy, or heuristic?

- Is ingestion batch, streaming, or episodic?
- What failure modes exist at the boundary?

Key Topics
- Schema-first vs schema-on-read
- Data contracts and assumptions
- Metadata as first-class input
- Multimodal ingestion (tables, text, PDFs)

Prompt Engineering Pattern
“Given this data source, list assumptions, failure modes, and what decisions break if this data is wrong.”

🧪 Exercise
Ask AI to design:
- an ingestion plan, and
- a validation checklist

for a noisy external dataset.

MODULE 2 — Data Cleaning: Noise, Bias, and Tradeoffs
Bias Shaping, Not Fixing Data
Why This Matters
AI views cleaning as **bias shaping**, not error removal.
Cleaning decisions materially alter model behavior and decision outcomes.

AI Decision Questions
- Is this error random, systematic, or structural?
- Does cleaning increase or destroy signal?
- What downstream behavior does this change?

Key Topics
- Missingness mechanisms (conceptual MCAR / MAR / MNAR)
- Outliers as signal vs defects
- Domain-informed vs statistical cleaning
- Reproducible cleaning logic

Prompt Engineering Pattern
"Propose multiple cleaning strategies, explain their assumptions, and identify when each would be dangerous."

🧪 Exercise
Have AI generate **two conflicting cleaning strategies** and defend each.

MODULE 3 — Feature Engineering: Representation as Power
How the World Is Shown to the Model
Why This Matters
To AI, features determine **what relationships are even visible**.
Feature engineering is a representational choice, not just column creation.

AI Decision Questions
- What relationships must the model see?
- What invariances should be preserved?
- What leakage risks exist?

Key Topics
- Raw vs derived features
- Temporal and hierarchical features
- Domain-informed transformations
- Feature documentation (why it exists)

Prompt Engineering Pattern
"Given this decision goal, what features would a human miss but a model would exploit — and vice versa?"

🧪 Exercise
Ask AI to propose features and explicitly label which ones are **risky**.

MODULE 4 — Modeling: How AI Chooses Models
Constraint Satisfaction, Not Popularity
Why This Matters

AI does not select models by trend — it selects them by **constraint alignment**.

AI Model Selection Criteria

- Interpretability vs performance
- Data volume vs model capacity
- Stability vs adaptivity
- Regulatory and stakeholder constraints

Key Topics

- Baselines as anchors
- Model families as tradeoff bundles
- When *not* to use ML
- Ensemble thinking (conceptual)

Prompt Engineering Pattern

"Recommend three model classes, rank them under these constraints, and explain why the top choice could still fail."

🧪 Exercise

Force AI to **reject a deep learning approach** and justify why.

MODULE 5 — Evaluation: Metrics, Thresholds, and Decision Proxies

From Scores to Actions

Why This Matters

Evaluation is not neutral — it determines **how the system behaves in the real world**.

AI Decision Questions

- What decision is this metric optimizing?
- Who benefits or loses if it's wrong?
- How stable is this metric over time?

Key Topics

- Metric alignment vs metric gaming
- Validation vs testing vs monitoring
- Error analysis as learning

- Scenario-based evaluation

Prompt Engineering Pattern
“Design an evaluation plan that would convince a skeptical stakeholder this model is safe to use.”

🧪 Exercise
Ask AI to propose metrics **and argue against them**.

MODULE 6 — End-to-End Pipeline Reasoning
Pipelines as Graphs with Feedback
Why This Matters
AI reasons about pipelines as **graphs**, not straight lines.

Key Topics
- Pipeline DAGs
- Human review checkpoints
- Safe vs dangerous automation
- Intermediate artifacts

Prompt Engineering Pattern
“Represent this pipeline as a DAG, label risk points, and identify where human review is mandatory.”

🧪 Exercise
Convert an existing notebook workflow into a **decision-aware pipeline graph**.

MODULE 7 — Human–AI Collaboration in Data Science
Co-Design, Not Automation
Why This Matters
The strongest systems are **co-designed** with AI, not delegated to it.

Key Topics
- AI as analyst vs reviewer vs implementer

- Iterative prompting loops
- Avoiding premature convergence
- Documenting AI-assisted decisions

🧪 Exercise
Conduct a mock **AI design review** where the model critiques its own pipeline.

MODULE 8 — Data Science Readiness for Production
Bridge Module
Why This Matters
This module hands off cleanly into **Production Foundations**.

Key Topics
- What production teams need from data science
- What must be frozen vs flexible
- Model cards, data cards, assumptions
- When to stop experimenting

🧪 Exercise
Produce a **one-page DS handoff document** suitable for production transition.

Capstone — AI-Native Data Science System
Build a system that:
- accepts messy real-world data
- makes assumptions explicit
- produces defensible outputs
- explains *why* decisions were made

Deliverables
- Decision statement
- Pipeline DAG
- Model tradeoff analysis
- Evaluation plan
- Production handoff summary

🎯 Lesson Plan Outcomes

After completing this lesson plan, you will be able to:

- Define data science precisely and defensibly
- Explain DS decisions in AI-native language
- Prompt AI differently at each pipeline stage
- Enter Genesis-style conversations already aligned
- Transition smoothly into production thinking

Why This Fits Cleanly Into Your Curriculum

- **CS Foundations** → how AI thinks
- **DS Foundations** → what AI decides
- **Data Contracts** → where risk enters
- **Production Foundations** → how decisions survive reality

This creates a **continuous cognitive ladder**, not disconnected skill sets.